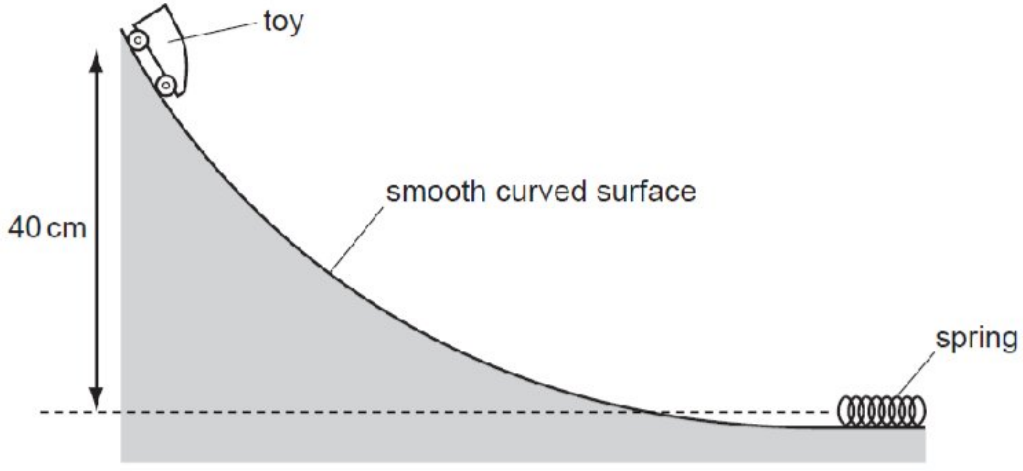
4 A toy, of mass 0.22 kg, is released from rest on a smooth curved surface as shown.



It slides down the curved surface through a vertical distance of 40 cm and strikes a spring of force constant 350 N m^{-1}, and compresses it a distance x before coming to rest momentarily.

Assuming no energy loss, what is the value of x?

A 4.9×10^{-3} m

B 7.2×10^{-3} m

C 2.2×10^{-2} m

D 7.0×10^{-2} m